

Gabrielle Rafert

WA Bee Atlas Collection and Identification Report - 2024

1 Your 2024 Collections

Gabrielle Rafert caught 3 bees across 1 county from May 12, 2024 to May 12, 2024 representing 2 unique taxa, including 2 unique species.

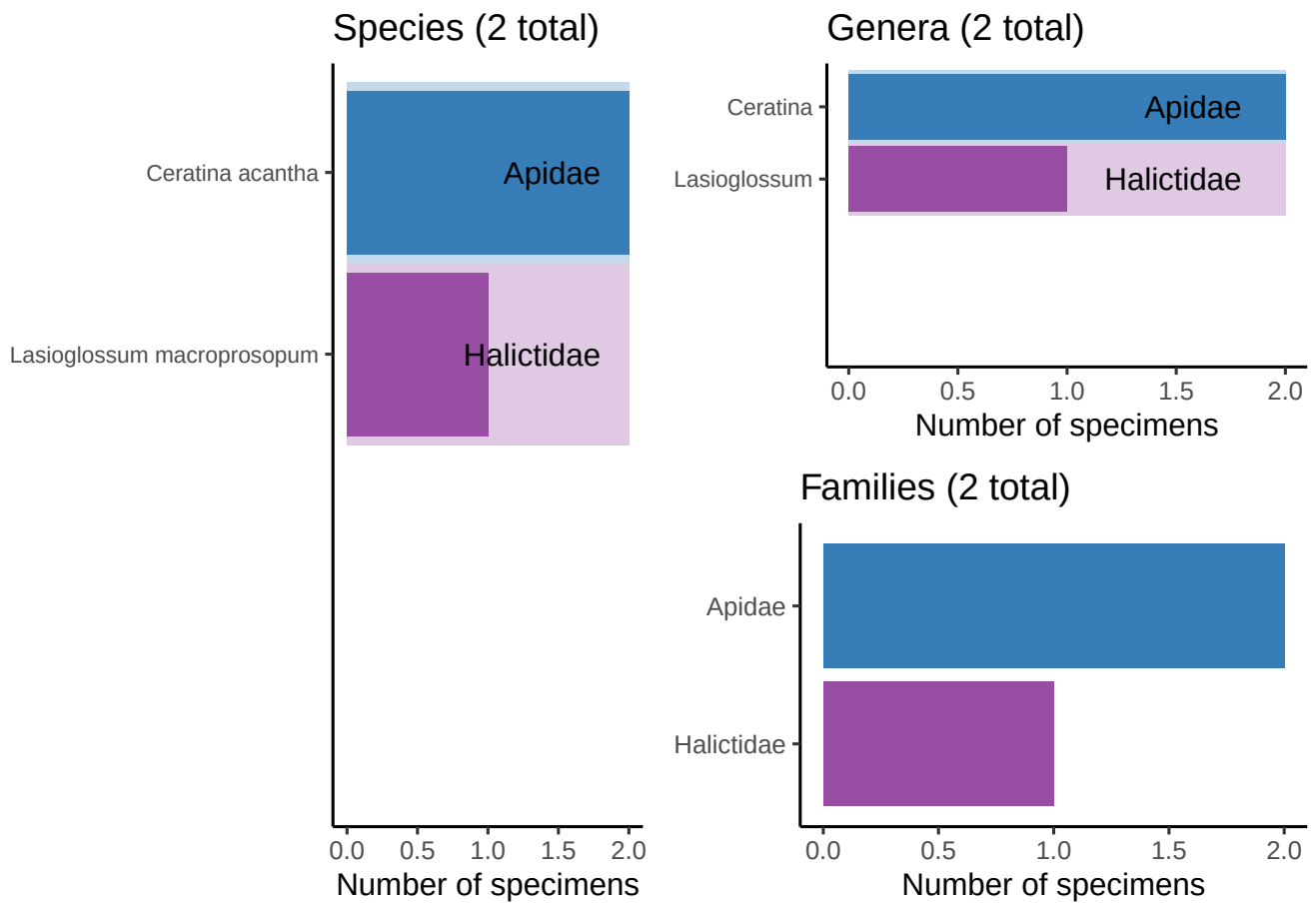


Figure 1: Bees caught by Gabrielle Rafert, broken down by species, genus, and family.

2 All Your Collections

Gabrielle Rafert caught 3 bees across 1 county from May 12, 2024 to May 12, 2024, representing 2 unique taxa, including 2 unique species.

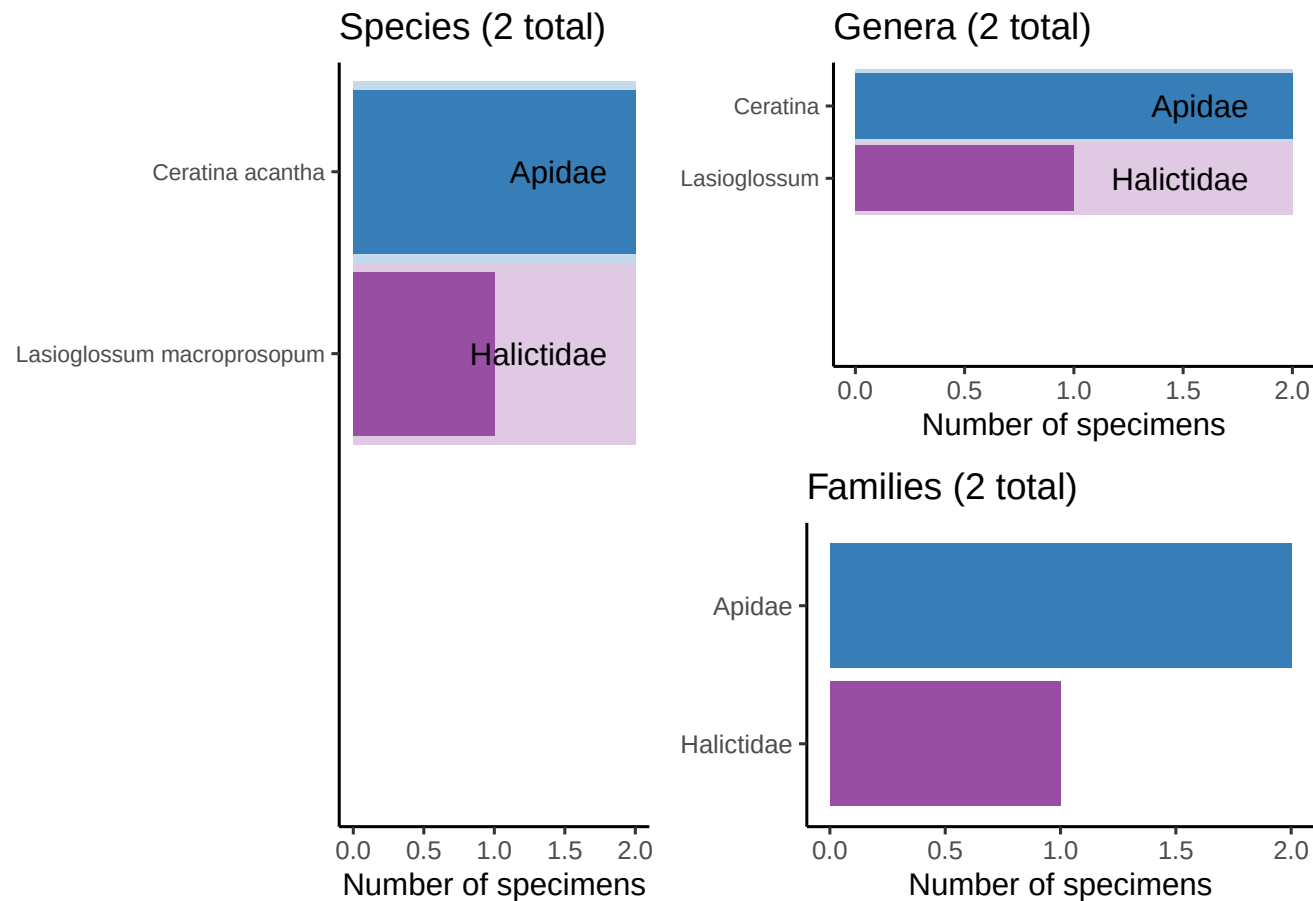


Figure 2: Bees caught by Gabrielle Rafert, broken down by species, genus, and family.

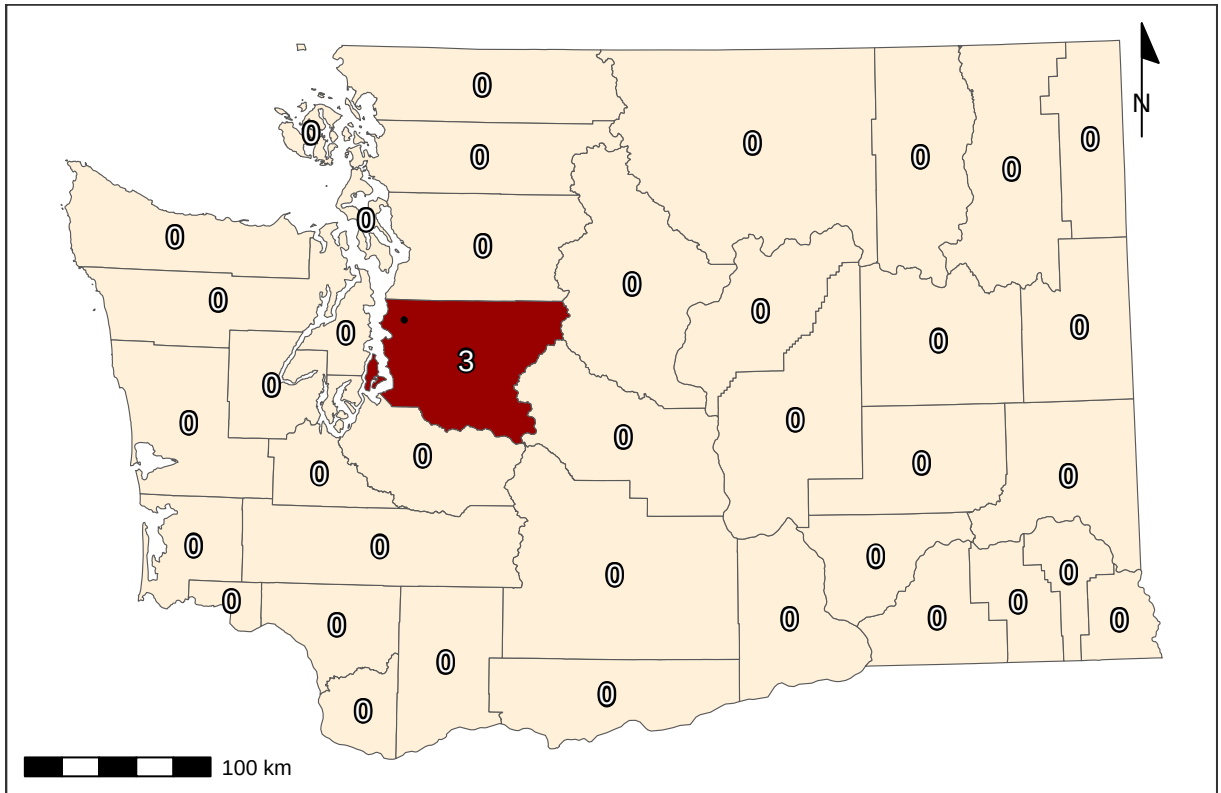


Figure 3: Bee catch locations for Gabrielle Rafert (within Washington), along with total catches per county.

Volunteers from the Washington Bee Atlas project caught 10006 bees across 35 counties from March 10, 2024 to November 14, 2024, representing 354 unique species and 43 unique genera. The **Nimble Net Kudos** (most specimens collected) goes to Karla Salp, Karen Wright, and Ingrid Carmean, who caught a total of 1689, 1005, and 957 specimens. The *positive* kind of **Darwin Award** (most species collected) goes to Karla Salp, Karen Wright, and Ellery Newcomer, who caught a total of 220, 167, and 153 unique species. Well done!

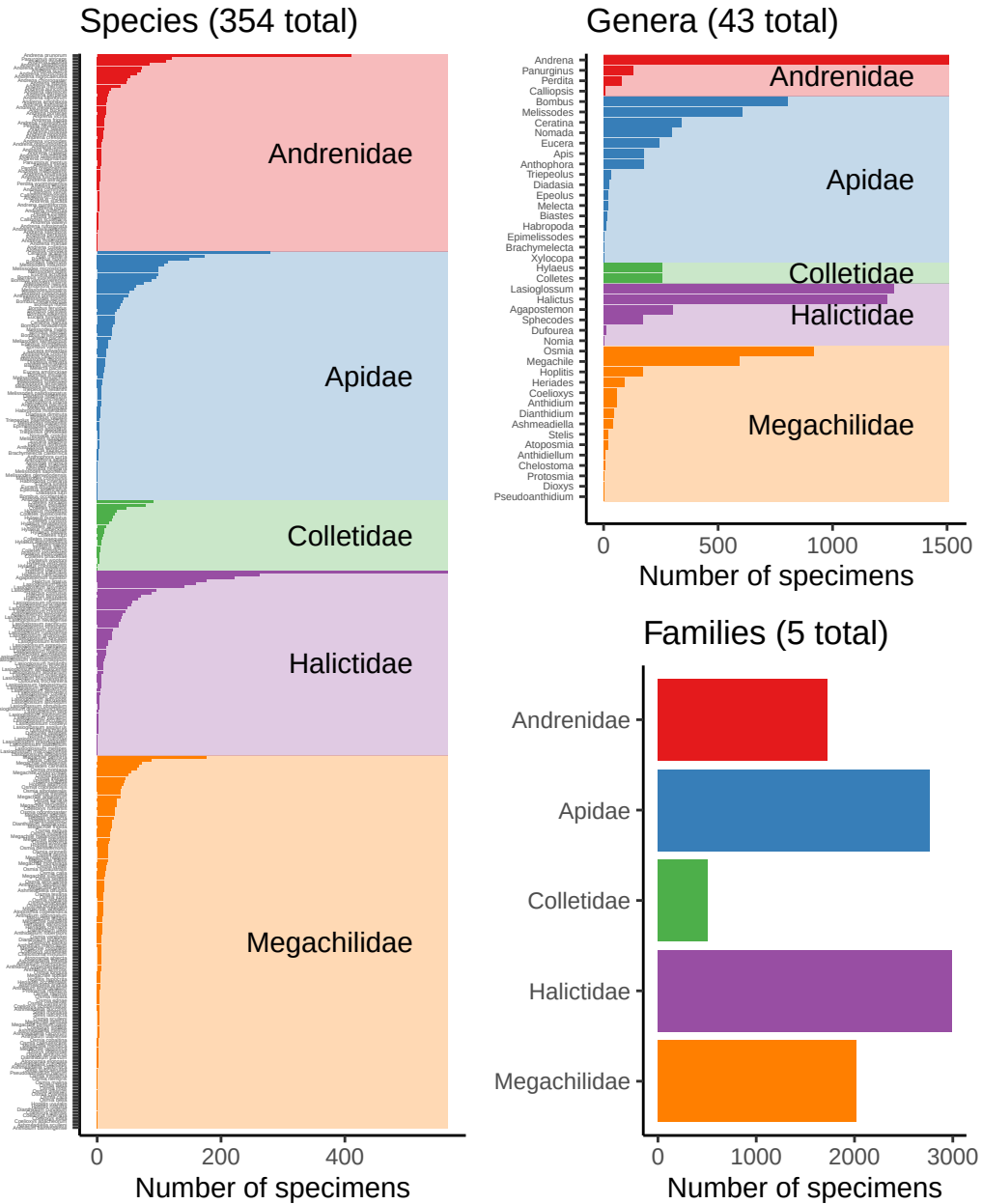


Figure 4: Bees caught by all volunteers, broken down by species, genus, and family.

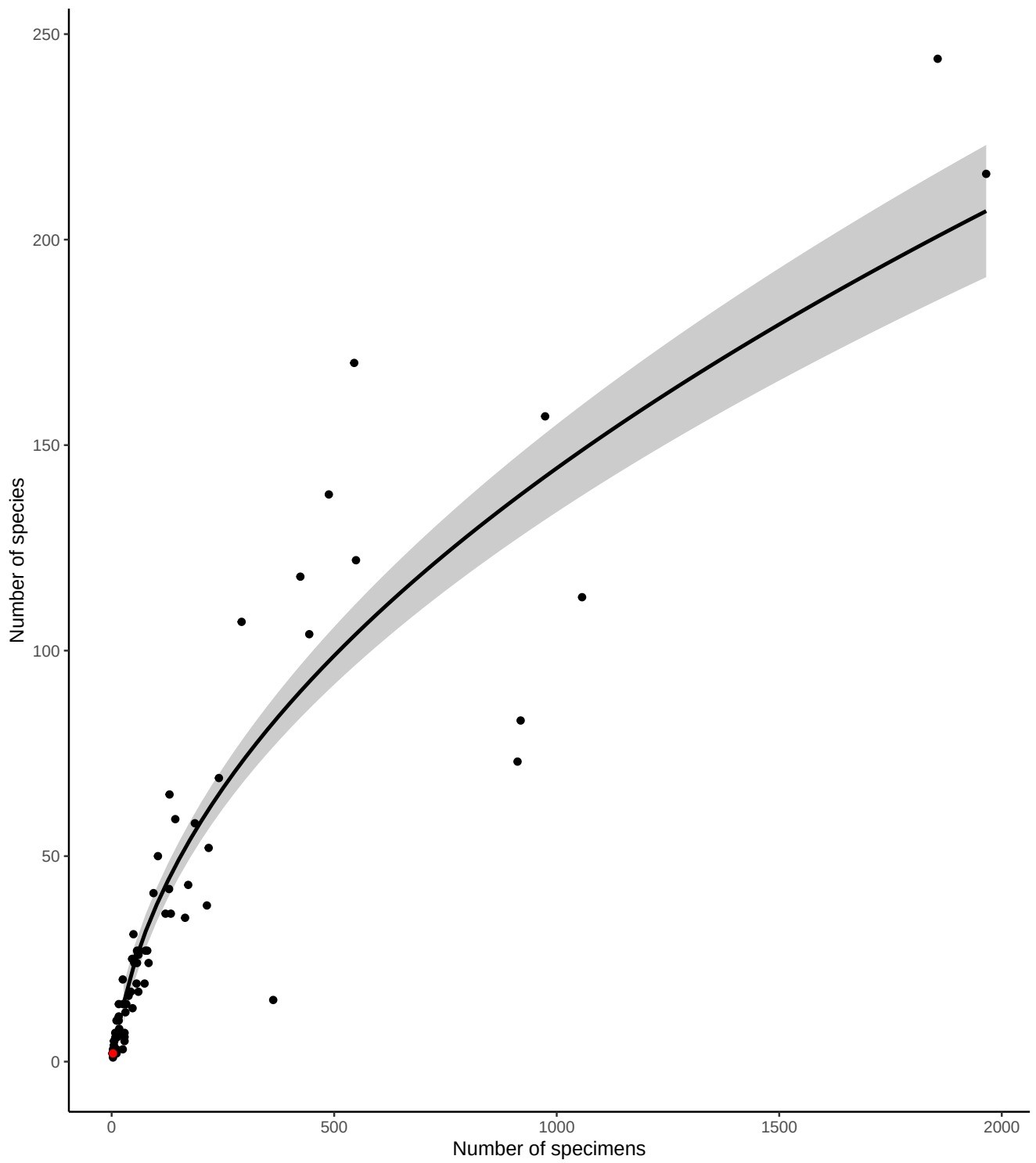


Figure 5: Number of bee specimens and unique bee species caught by all volunteers, with your effort shown in red. This graph should give you an idea of how many specimens you would need to catch to begin seeing rarer bee species.

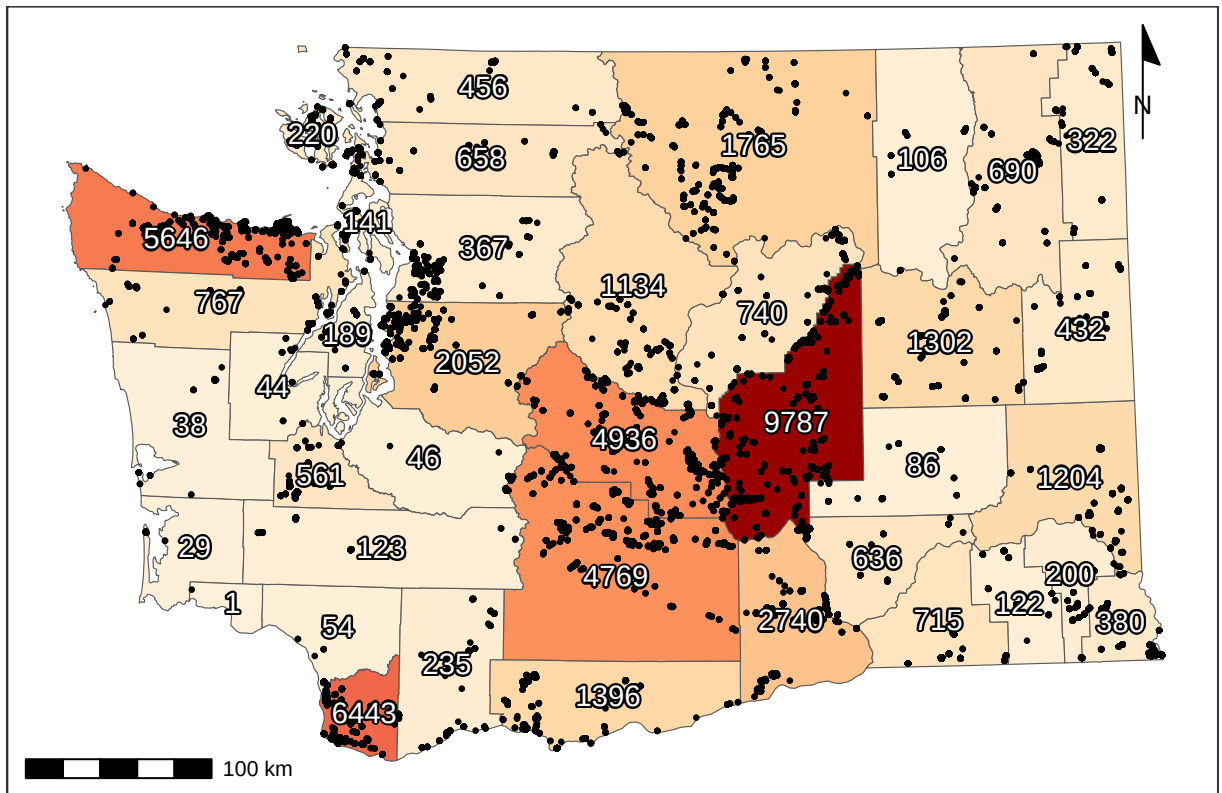


Figure 6: Total specimens caught per county, along with catch location of each specimen (black dots). For genus- and species-specific information for each county, see Tables 3 and 4.

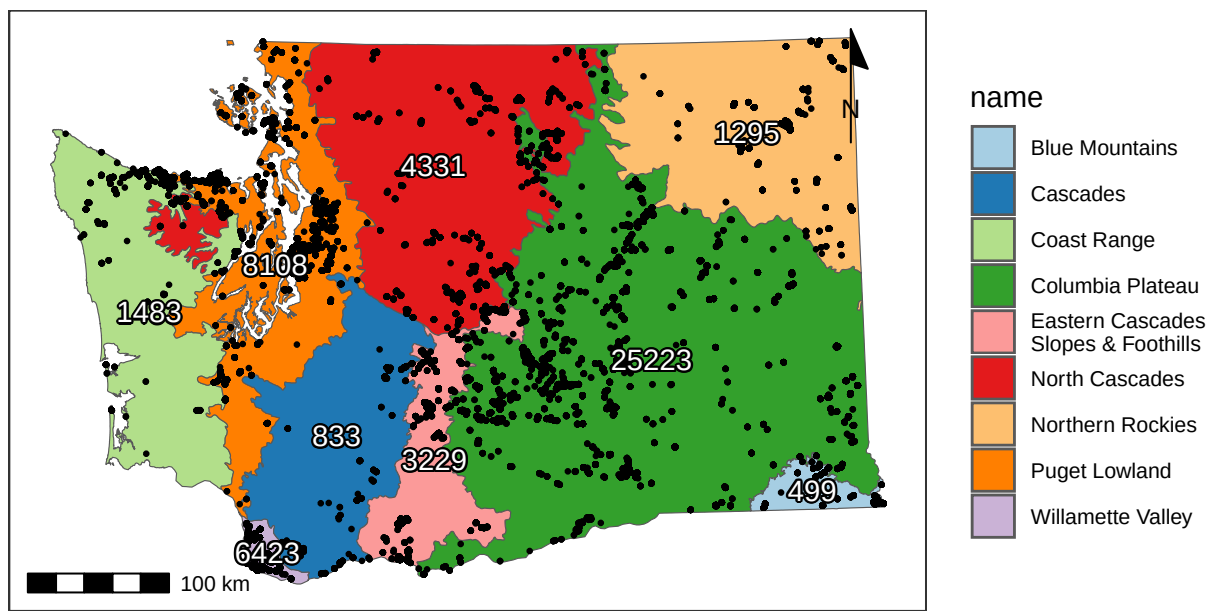


Figure 7: Total catches per ecoregion (Level III), along with catch location of each specimen (black dots).

4 Flight Phenology

4.1 West of (and including) the Cascade Mountains

Most bees (90%) were caught between April 16 and September 01, but the peak of season (50% of specimens) was from June 11 to July 27.

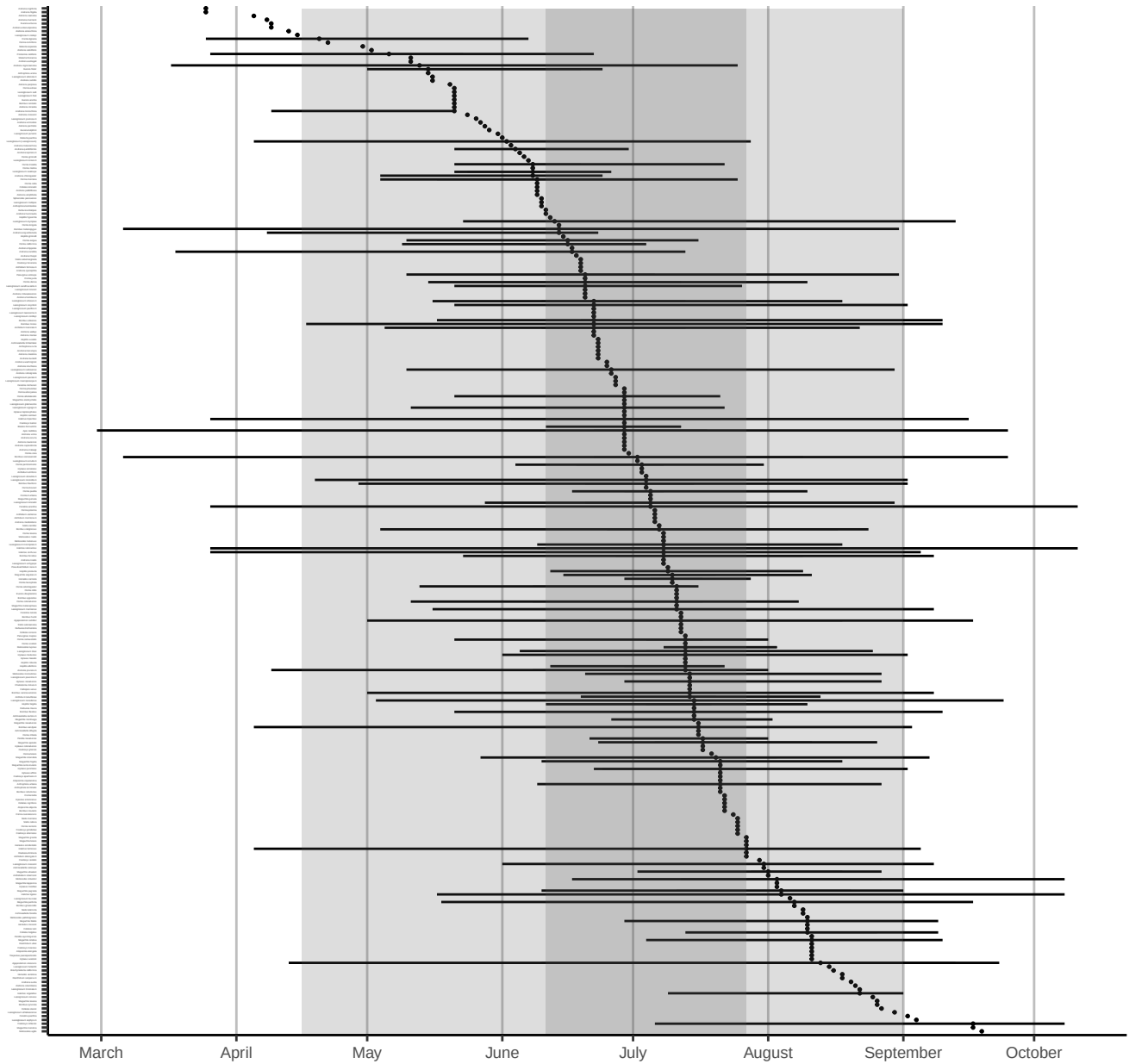


Figure 8: Phenology plot for all bee species caught in or West of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

4.2 East of the Cascade Mountains

Most bees (90%) were caught between April 14 and September 18, but the peak of season (50% of specimens) was from May 15 to July 16.

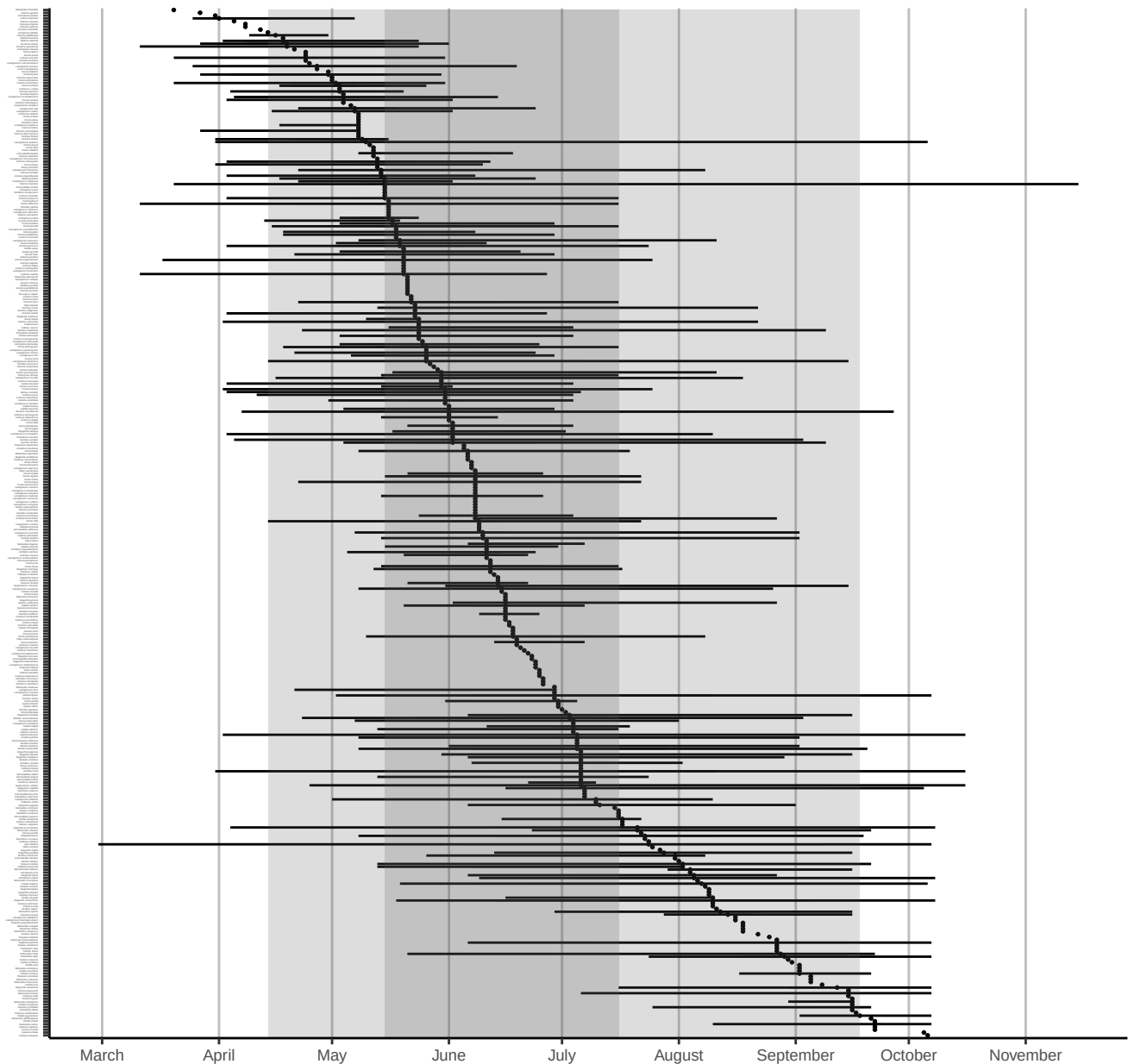


Figure 9: Phenology plot for all bee species caught east of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

5 Plant genera

Volunteers collected specimens from a total of 371 unique flower genera, with most volunteers sampling from 12 flower genera (median value). The **Flower Power Kudos** (most sampled flower genera) goes to Karla Salp, Karen Wright, and Nancy Carlson, who collected bees from a total of 170, 155, and 118 genera of flowers. Well done!

The flower genera that had the most specimens caught on them were *Grindelia*, *Ericameria*, and *Lomatium*, which yielded a total of 1095, 731, and 717 specimens. The flower genera that were popular with the most species of bees were *Erigeron*, *Phacelia*, and *Cirsium*, hosting a total of 101, 98, and 91 unique bee species. See Tables 1 and 2 for more details.

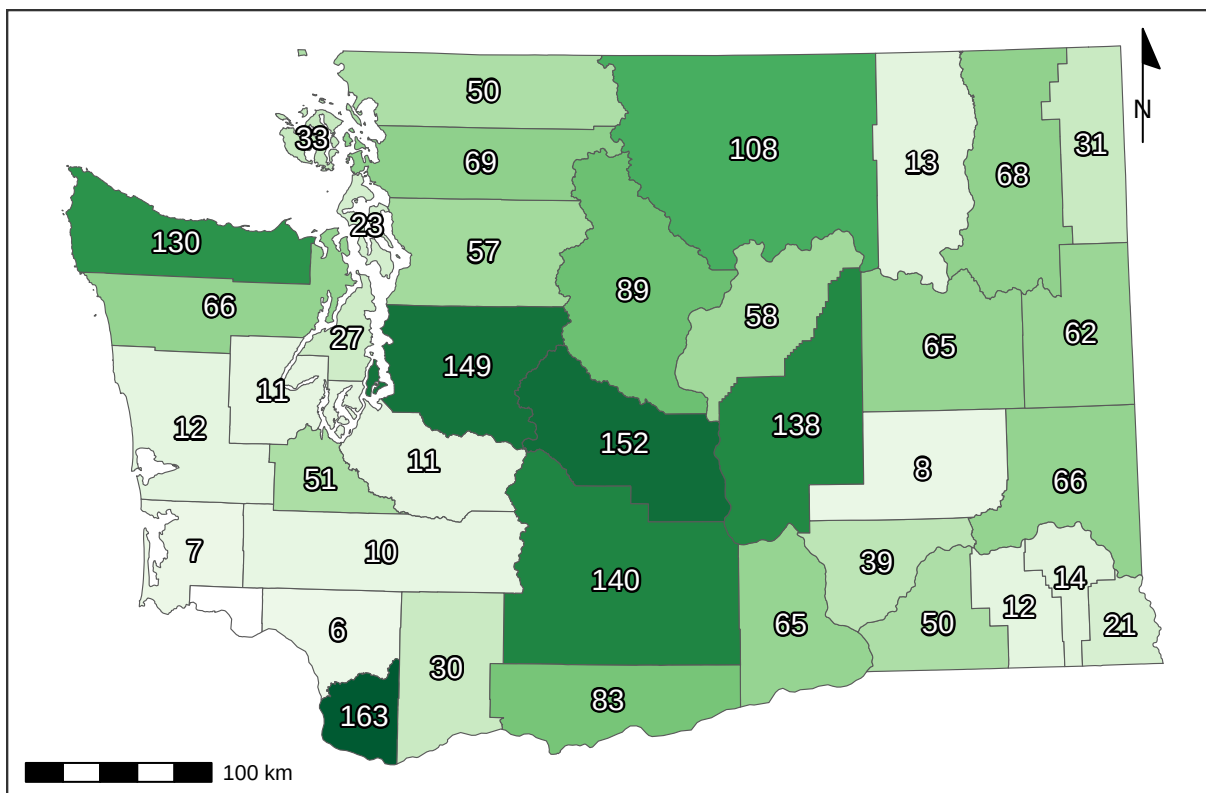


Figure 10: Recorded number of flower genera per county.

Table 1: Number of bee specimens collected from each plant genus. Plants with few records are great targets for future sampling.

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Grindelia</i>	1095	<i>Amsinckia</i>	97	<i>Delphinium</i>	35	<i>Angelica</i>	18	<i>Armeria</i>	8	<i>Anthemis</i>	4	<i>Abelia</i>	2
<i>Ericameria</i>	731	<i>Berberis</i>	97	<i>Malva</i>	35	<i>Acmispon</i>	17	<i>Coriandrum</i>	8	<i>Artemisia</i>	3	<i>Androsace</i>	2
<i>Lomatium</i>	717	<i>Dieteria</i>	94	<i>Nemophila</i>	34	<i>Euphorbia</i>	17	<i>Cuphea</i>	8	<i>Asparagus</i>	3	<i>Arenaria</i>	2
<i>Hypochaeris</i>	655	<i>Eschscholzia</i>	94	<i>Syringa</i>	34	<i>Cacaliopsis</i>	16	<i>Cytisus</i>	8	<i>Barbarea</i>	3	<i>Cardamine</i>	2
<i>Phacelia</i>	624	<i>Fragaria</i>	93	<i>Drymocalis</i>	33	<i>Lewisia</i>	16	<i>Dipsacus</i>	8	<i>Dimorphotheca</i>	3	<i>Corydalis</i>	2
<i>Helianthus</i>	490	<i>Origanum</i>	91	<i>Descraineria</i>	32	<i>Campanula</i>	15	<i>Downingia</i>	8	<i>Eremogone</i>	3	<i>Cota</i>	2
<i>Balsamorhiza</i>	471	<i>Clarkia</i>	89	<i>Malus</i>	32	<i>Draba</i>	15	<i>Hesperochiron</i>	8	<i>Galium</i>	3	<i>Crocasmia</i>	2
<i>Cirsium</i>	450	<i>Medicago</i>	88	<i>Crataegus</i>	31	<i>Hackelia</i>	15	<i>Aquilegia</i>	7	<i>Gayophytum</i>	3	<i>Crocus</i>	2
<i>Erigeron</i>	446	<i>Ceanothus</i>	85	<i>Doellingeria</i>	31	<i>Heterotheca</i>	15	<i>Erica</i>	7	<i>Geum</i>	3	<i>Achlys</i>	1
<i>Solidago</i>	429	<i>Daucus</i>	83	<i>Myosotis</i>	31	<i>Lonicera</i>	15	<i>Fritillaria</i>	7	<i>Hibiscus</i>	3	<i>Adelphia</i>	1
<i>Penstemon</i>	414	<i>Sphaeralcea</i>	82	<i>Tunacetum</i>	31	<i>Oenothera</i>	15	<i>Viburnum</i>	7	<i>Impatiens</i>	3	<i>Albizia</i>	1
<i>Leucanthemum</i>	413	<i>Lathyrus</i>	78	<i>Bellis</i>	30	<i>Perideridia</i>	15	<i>Amelanchier</i>	6	<i>Lapsana</i>	3	<i>Anthozanthum</i>	1
<i>Sisymbrium</i>	408	<i>Pediocactus</i>	78	<i>Cleome</i>	30	<i>Ratibida</i>	15	<i>Anchusa</i>	6	<i>Liatris</i>	3	<i>Antirrhinum</i>	1
<i>Eriogonum</i>	381	<i>Epilobium</i>	77	<i>Hieracium</i>	30	<i>Antennaria</i>	14	<i>Cynara</i>	6	<i>Lobularia</i>	3	<i>Bellardia</i>	1
<i>Lupinus</i>	372	<i>Rudbeckia</i>	77	<i>Rorippa</i>	30	<i>Crocidium</i>	14	<i>Cynoglossum</i>	6	<i>Nothochelone</i>	3	<i>Boechera</i>	1
<i>Potentilla</i>	320	<i>Monardella</i>	74	<i>Cerastium</i>	29	<i>Frangula</i>	14	<i>Limnanthes</i>	6	<i>Oenanthe</i>	3	<i>Buglossoides</i>	1
<i>Rubus</i>	285	<i>Physocarpus</i>	71	<i>Collinsia</i>	29	<i>Hydrophyllum</i>	14	<i>Luetkea</i>	6	<i>Oreocarya</i>	3	<i>Calystegia</i>	1
<i>Rosa</i>	269	<i>Camassia</i>	70	<i>Lactuca</i>	29	<i>Plantago</i>	14	<i>Microseris</i>	6	<i>Physaria</i>	3	<i>Capsella</i>	1
<i>Symphoricarpos</i>	265	<i>Spiraea</i>	70	<i>Olaynia</i>	29	<i>Borago</i>	13	<i>Nigella</i>	6	<i>Pieris</i>	3	<i>Carduus</i>	1
<i>Taraxacum</i>	258	<i>Calochortus</i>	69	<i>Clematis</i>	28	<i>Heleum</i>	13	<i>Onopordum</i>	6	<i>Primula</i>	3	<i>Cercis</i>	1
<i>Madia</i>	256	<i>Nepeta</i>	68	<i>Acer</i>	27	<i>Bistorta</i>	12	<i>Phoenixaulis</i>	6	<i>Pseudognaphalium</i>	3	<i>Chamaecrista</i>	1
<i>Holodiscus</i>	247	<i>Coreopsis</i>	67	<i>Canadanthus</i>	27	<i>Cucurbita</i>	12	<i>Robinia</i>	6	<i>Sagittaria</i>	3	<i>Comandra</i>	1
<i>Achillea</i>	246	<i>Agastache</i>	66	<i>Sonchus</i>	27	<i>Heuchera</i>	12	<i>Ageratina</i>	5	<i>Saponaria</i>	3	<i>Cotoneaster</i>	1
<i>Asclepias</i>	228	<i>Hypericum</i>	60	<i>Sphaerophysa</i>	26	<i>Persicaria</i>	12	<i>Ajuga</i>	5	<i>Silene</i>	3	<i>Cryptantha</i>	1
<i>Lotus</i>	224	<i>Convolvulus</i>	59	<i>Linum</i>	25	<i>Spergularia</i>	12	<i>Ammi</i>	5	<i>Tellima</i>	3	<i>Cucumis</i>	1
<i>Trifolium</i>	218	<i>Cichorium</i>	55	<i>Wyethia</i>	25	<i>Vitex</i>	12	<i>Arctostaphylos</i>	5	<i>Thalictrum</i>	3	<i>Dicentra</i>	1
<i>Crepis</i>	217	<i>Veronica</i>	55	<i>Aruncus</i>	24	<i>Centaurium</i>	11	<i>Bassia</i>	5	<i>Thermopsis</i>	3	<i>Echinops</i>	1
<i>Brassica</i>	213	<i>Arnica</i>	54	<i>Chondrilla</i>	24	<i>Chorispura</i>	11	<i>Cymopterus</i>	5	<i>Visnaga</i>	3	<i>Erythronium</i>	1
<i>Centauria</i>	210	<i>Digitalis</i>	54	<i>Lithospermum</i>	24	<i>Cosmos</i>	11	<i>Helopsis</i>	5	<i>Dalea</i>	2	<i>Gnaphalium</i>	1
<i>Prunus</i>	202	<i>Linaria</i>	53	<i>Anethum</i>	23	<i>Dahlia</i>	11	<i>Hydrangea</i>	5	<i>Eurybia</i>	2	<i>Holcus</i>	1
<i>Salix</i>	200	<i>Chaenactis</i>	52	<i>Purshia</i>	23	<i>Micranthes</i>	11	<i>Melissa</i>	5	<i>Fagopyrum</i>	2	<i>Hylotelephium</i>	1
<i>Ribes</i>	188	<i>Nothocalais</i>	51	<i>Hieracium</i>	22	<i>Pseudotsuga</i>	11	<i>Mycelis</i>	5	<i>Fothergilla</i>	2	<i>Hymenopappus</i>	1
<i>Ranunculus</i>	184	<i>Sedum</i>	51	<i>Mentzelia</i>	22	<i>Pyrocampa</i>	11	<i>Narcissus</i>	5	<i>Hesperis</i>	2	<i>Hyssopus</i>	1
<i>Anaphalis</i>	179	<i>Euthamia</i>	47	<i>Thelypodium</i>	22	<i>Valerianella</i>	11	<i>Petasites</i>	5	<i>Ionactis</i>	2	<i>Ilex</i>	1
<i>Senecio</i>	176	<i>Cornus</i>	44	<i>Triteleia</i>	22	<i>Asperugo</i>	10	<i>Phedimus</i>	5	<i>Kalmia</i>	2	<i>Ipomopsis</i>	1
<i>Chrysothamnus</i>	175	<i>Helianthella</i>	44	<i>Papaver</i>	21	<i>Berteroa</i>	10	<i>Sorbus</i>	5	<i>Ladecania</i>	2	<i>Isocoma</i>	1
<i>Melilotus</i>	167	<i>Lavandula</i>	44	<i>Polemonium</i>	21	<i>Erodium</i>	10	<i>Argentina</i>	4	<i>Ligusticum</i>	2	<i>Lamprocapnos</i>	1
<i>Gaillardia</i>	154	<i>Mentha</i>	44	<i>Erythranthe</i>	20	<i>Lamium</i>	10	<i>Cistus</i>	4	<i>Lycium</i>	2	<i>Lomelosia</i>	1
<i>Chamaenerion</i>	147	<i>Plagiobothrys</i>	44	<i>Frasera</i>	20	<i>Lygodesmia</i>	10	<i>Dichelostemma</i>	4	<i>Mertensia</i>	2	<i>Lycopus</i>	1
<i>Philadelphus</i>	146	<i>Plectritis</i>	43	<i>Iris</i>	20	<i>Marrubium</i>	10	<i>Echinacea</i>	4	<i>Montia</i>	2	<i>Matricaria</i>	1
<i>Albium</i>	144	<i>Pastinaca</i>	42	<i>Oemeria</i>	20	<i>Monarda</i>	10	<i>Elaeagnus</i>	4	<i>Nasturtium</i>	2	<i>Muscari</i>	1
<i>Lepidium</i>	134	<i>Rhododendron</i>	42	<i>Phlox</i>	20	<i>Solanum</i>	10	<i>Eutrochium</i>	4	<i>Noccea</i>	2	<i>Nicotiana</i>	1
<i>Symphoricarpos</i>	132	<i>Vaccinium</i>	42	<i>Rhus</i>	20	<i>Tragopogon</i>	10	<i>Glycyrrhiza</i>	4	<i>Osmia</i>	2	<i>Oxalis</i>	1
<i>Vicia</i>	120	<i>Lithophragma</i>	40	<i>Toxicoscordion</i>	20	<i>Fallopia</i>	9	<i>Iberis</i>	4	<i>Ozomelis</i>	2	<i>Phalaris</i>	1
<i>Cilia</i>	119	<i>Erysimum</i>	39	<i>Abronia</i>	19	<i>Pyrrus</i>	9	<i>Ivesia</i>	4	<i>Pedicularis</i>	2	<i>Pisum</i>	1
<i>Salvia</i>	117	<i>Nestodus</i>	39	<i>Anthriscus</i>	19	<i>Rhaphiolepis</i>	9	<i>Machaeranthera</i>	4	<i>Petroselinum</i>	2	<i>Polygonum</i>	1
<i>Jacobaea</i>	116	<i>Calendula</i>	38	<i>Castilleja</i>	19	<i>Tripleurospermum</i>	9	<i>Microsteris</i>	4	<i>Pilosella</i>	2	<i>Salsola</i>	1
<i>Eriophyllum</i>	111	<i>Dasiphora</i>	38	<i>Gaultheria</i>	19	<i>Leontodon</i>	8	<i>Mutarda</i>	4	<i>Poa</i>	2	<i>Sambucus</i>	1
<i>Astragalus</i>	104	<i>Sidalcea</i>	38	<i>Thymus</i>	19	<i>Levistica</i>	8	<i>Opuntia</i>	4	<i>Pyraecantha</i>	2	<i>Santolina</i>	1
<i>Apocynum</i>	102	<i>Cakile</i>	37	<i>Collomia</i>	18	<i>Paeonia</i>	8	<i>Petrosedum</i>	4	<i>Satureja</i>	2	<i>Sisyrinchium</i>	1
<i>Geranium</i>	99	<i>Claytonia</i>	37	<i>Oxytropis</i>	18	<i>Scrophularia</i>	8	<i>Stellaria</i>	4	<i>Symphytum</i>	2	<i>Urtica</i>	1
<i>Ilamna</i>	99	<i>Prunella</i>	35	<i>Stephanomeria</i>	18	<i>Stachys</i>	8	<i>Tolmiea</i>	4	<i>Viola</i>	2	<i>Veratrum</i>	1
<i>Lythrum</i>	97	<i>Rhaponticum</i>	35	<i>Tagetes</i>	18	<i>Valeriana</i>	8	<i>Verbena</i>	4	<i>Zinnia</i>	2	<i>Xerophyllum</i>	1

Table 2: Number of bee species collected from each plant genus

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Erigeron</i>	101	<i>Coreopsis</i>	31	<i>Bellis</i>	15	<i>Antennaria</i>	8	<i>Ageratina</i>	4	<i>Argentina</i>	2	<i>Abelia</i>	1
<i>Phacelia</i>	98	<i>Agastache</i>	29	<i>Crataegus</i>	14	<i>Aruncus</i>	8	<i>Ajuga</i>	4	<i>Artemisia</i>	2	<i>Achlys</i>	1
<i>Cirsium</i>	91	<i>Eschscholzia</i>	29	<i>Heracleum</i>	14	<i>Berteroa</i>	8	<i>Ammi</i>	4	<i>Borago</i>	2	<i>Adelphia</i>	1
<i>Grindelia</i>	91	<i>Monardella</i>	29	<i>Hieracium</i>	14	<i>Cacaliopsis</i>	8	<i>Cistus</i>	4	<i>Cardamine</i>	2	<i>Albizia</i>	1
<i>Eriogonum</i>	86	<i>Berberis</i>	28	<i>Malus</i>	14	<i>Canadanthus</i>	8	<i>Cosmos</i>	4	<i>Cota</i>	2	<i>Androsace</i>	1
<i>Penstemon</i>	85	<i>Chrysothamnus</i>	28	<i>Malva</i>	14	<i>Cleome</i>	8	<i>Cuphea</i>	4	<i>Crocosmia</i>	2	<i>Anthozanthum</i>	1
<i>Solidago</i>	82	<i>Convolvulus</i>	28	<i>Tanacetum</i>	14	<i>Erysimum</i>	8	<i>Cymopterus</i>	4	<i>Dalea</i>	2	<i>Antirrhinum</i>	1
<i>Crepis</i>	75	<i>Dieteria</i>	28	<i>Lithospermum</i>	13	<i>Heuchera</i>	8	<i>Dahlia</i>	4	<i>Eurybia</i>	2	<i>Arenaria</i>	1
<i>Hypochaeris</i>	75	<i>Fragaria</i>	28	<i>Parashia</i>	13	<i>Hydrophyllum</i>	8	<i>Dichlostemma</i>	4	<i>Galium</i>	2	<i>Bellardia</i>	1
<i>Lupinus</i>	75	<i>Calochortus</i>	27	<i>Sonchus</i>	13	<i>Anethum</i>	7	<i>Echinacea</i>	4	<i>Geum</i>	2	<i>Boechera</i>	1
<i>Leucanthemum</i>	69	<i>Chaenactis</i>	27	<i>Collomia</i>	12	<i>Anthriscus</i>	7	<i>Eutrochium</i>	4	<i>Glycyrrhiza</i>	2	<i>Buglossoides</i>	1
<i>Rosa</i>	69	<i>Lathyrus</i>	27	<i>Doellingeria</i>	12	<i>Campanula</i>	7	<i>Frangula</i>	4	<i>Hesperis</i>	2	<i>Calystegia</i>	1
<i>Potentilla</i>	67	<i>Madia</i>	26	<i>Hackelia</i>	12	<i>Centaurium</i>	7	<i>Fritillaria</i>	4	<i>Iberis</i>	2	<i>Capsella</i>	1
<i>Sisymbrium</i>	64	<i>Medicago</i>	26	<i>Lactuca</i>	12	<i>Helenium</i>	7	<i>Heliopsis</i>	4	<i>Impatiens</i>	2	<i>Carduus</i>	1
<i>Achillea</i>	63	<i>Hypericum</i>	25	<i>Plantago</i>	12	<i>Heterotheca</i>	7	<i>Hydrangea</i>	4	<i>Ionactis</i>	2	<i>Cercis</i>	1
<i>Ericameria</i>	63	<i>Rudbeckia</i>	25	<i>Syringa</i>	12	<i>Micranthes</i>	7	<i>Ivesia</i>	4	<i>Kalmia</i>	2	<i>Chamaecrista</i>	1
<i>Lomatium</i>	63	<i>Arnica</i>	24	<i>Triteleia</i>	12	<i>Perideridia</i>	7	<i>Machaeranthera</i>	4	<i>Levisticum</i>	2	<i>Comandra</i>	1
<i>Balsamorhiza</i>	62	<i>Lythrum</i>	24	<i>Vaccinium</i>	12	<i>Rorippa</i>	7	<i>Microsteris</i>	4	<i>Liatris</i>	2	<i>Corydalis</i>	1
<i>Rubus</i>	62	<i>Nepeta</i>	24	<i>Wyethia</i>	12	<i>Stephanomeria</i>	7	<i>Mycelis</i>	4	<i>Ligusticum</i>	2	<i>Cotoneaster</i>	1
<i>Centauria</i>	60	<i>Cichorium</i>	23	<i>Collinsia</i>	11	<i>Abronia</i>	6	<i>Narcissus</i>	4	<i>Limnanthes</i>	2	<i>Crocus</i>	1
<i>Chamaenerion</i>	59	<i>Clematis</i>	23	<i>Lonicera</i>	11	<i>Amelanchier</i>	6	<i>Oenleria</i>	4	<i>Labularia</i>	2	<i>Cryptantha</i>	1
<i>Allium</i>	58	<i>Epilobium</i>	23	<i>Mentzelia</i>	11	<i>Aquilegia</i>	6	<i>Onopordum</i>	4	<i>Lycium</i>	2	<i>Cucumis</i>	1
<i>Senecio</i>	51	<i>Claytonia</i>	22	<i>Olepnium</i>	11	<i>Armeria</i>	6	<i>Phedimus</i>	4	<i>Montia</i>	2	<i>Dicentra</i>	1
<i>Helianthus</i>	50	<i>Euthamia</i>	22	<i>Sphaerophysa</i>	11	<i>Castilleja</i>	6	<i>Stachys</i>	4	<i>Nasturtium</i>	2	<i>Downingia</i>	1
<i>Holodiscus</i>	50	<i>Ilama</i>	21	<i>Thelypodium</i>	11	<i>Crocidium</i>	6	<i>Toxicoscordion</i>	4	<i>Nigella</i>	2	<i>Echinops</i>	1
<i>Melilotus</i>	50	<i>Linaria</i>	21	<i>Thymus</i>	11	<i>Cucurbita</i>	6	<i>Anthemis</i>	3	<i>Nothochelone</i>	2	<i>Erythronium</i>	1
<i>Prunus</i>	50	<i>Nothocalais</i>	21	<i>Acer</i>	10	<i>Erica</i>	6	<i>Arctostaphylos</i>	3	<i>Opuntia</i>	2	<i>Fagopyrum</i>	1
<i>Trifolium</i>	50	<i>Sedum</i>	21	<i>Chondrilla</i>	10	<i>Fullopia</i>	6	<i>Asparagus</i>	3	<i>Osmia</i>	2	<i>Fothergilla</i>	1
<i>Lotus</i>	49	<i>Philadelphus</i>	20	<i>Descurainia</i>	10	<i>Hesperochiron</i>	6	<i>Barbarea</i>	3	<i>Oxytropis</i>	2	<i>Gnaphalium</i>	1
<i>Taraxacum</i>	49	<i>Physocarpus</i>	20	<i>Digitalis</i>	10	<i>Leontodon</i>	6	<i>Bassia</i>	3	<i>Pedicularis</i>	2	<i>Holcus</i>	1
<i>Anaphalis</i>	45	<i>Sphaeralcea</i>	20	<i>Euphorbia</i>	10	<i>Monarda</i>	6	<i>Cynara</i>	3	<i>Petroselinum</i>	2	<i>Hylotelephium</i>	1
<i>Gaillardia</i>	45	<i>Camassia</i>	19	<i>Lewisia</i>	10	<i>Persicaria</i>	6	<i>Dimorphotheca</i>	3	<i>Physaria</i>	2	<i>Hymenopappus</i>	1
<i>Salvia</i>	45	<i>Cornus</i>	19	<i>Oenothera</i>	10	<i>Rhaphiolepis</i>	6	<i>Elaeagnus</i>	3	<i>Pieris</i>	2	<i>Hyssopus</i>	1
<i>Brassica</i>	43	<i>Plectritis</i>	19	<i>Plagiobothrys</i>	10	<i>Spergularia</i>	6	<i>Eremogone</i>	3	<i>Pyraecantha</i>	2	<i>Ilex</i>	1
<i>Ranunculus</i>	43	<i>Veronica</i>	19	<i>Polemonium</i>	10	<i>Tripleurospermum</i>	6	<i>Gayophytum</i>	3	<i>Stellaria</i>	2	<i>Ipomopsis</i>	1
<i>Symphotrichum</i>	43	<i>Daucus</i>	18	<i>Pseudotsuga</i>	10	<i>Valerianella</i>	6	<i>Hibiscus</i>	3	<i>Symphytum</i>	2	<i>Isocoma</i>	1
<i>Vicia</i>	43	<i>Delphinium</i>	18	<i>Acmispon</i>	9	<i>Anchusa</i>	5	<i>Lamium</i>	3	<i>Thermopsis</i>	2	<i>Ladecania</i>	1
<i>Lepidium</i>	42	<i>Drymocallis</i>	18	<i>Angelica</i>	9	<i>Asperugo</i>	5	<i>Lapsana</i>	3	<i>Viola</i>	2	<i>Lamprocapnos</i>	1
<i>Salix</i>	38	<i>Mentha</i>	18	<i>Bistorta</i>	9	<i>Coriandrum</i>	5	<i>Luetkea</i>	3	<i>Polygonum</i>	1	<i>Lomelosia</i>	1
<i>Symphoricarpos</i>	38	<i>Nastotus</i>	18	<i>Cerastium</i>	9	<i>Cynoglossum</i>	5	<i>Mutarda</i>	3	<i>Salsola</i>	1	<i>Lycopus</i>	1
<i>Asclepias</i>	37	<i>Prunella</i>	18	<i>Chorispora</i>	9	<i>Cytisus</i>	5	<i>Oenanthe</i>	3	<i>Sambucus</i>	1	<i>Matricaria</i>	1
<i>Amsinckia</i>	35	<i>Erythranthe</i>	17	<i>Erodium</i>	9	<i>Dipsacus</i>	5	<i>Oreocarya</i>	3	<i>Santolina</i>	1	<i>Melissa</i>	1
<i>Apocynum</i>	35	<i>Helianthella</i>	17	<i>Frasera</i>	9	<i>Draba</i>	5	<i>Phoenicautis</i>	3	<i>Satureja</i>	1	<i>Mertensia</i>	1
<i>Eriophyllum</i>	35	<i>Linum</i>	17	<i>Iris</i>	9	<i>Gaultheria</i>	5	<i>Primula</i>	3	<i>Sisyrinchium</i>	1	<i>Muscari</i>	1
<i>Geranium</i>	34	<i>Rhaponticum</i>	17	<i>Nemophila</i>	9	<i>Microseris</i>	5	<i>Pseudognaphalium</i>	3	<i>Tellima</i>	1	<i>Nicotiana</i>	1
<i>Jacobaea</i>	34	<i>Cakile</i>	16	<i>Pastinaca</i>	9	<i>Paeonia</i>	5	<i>Pyrus</i>	3	<i>Thalictrum</i>	1	<i>Noccea</i>	1
<i>Ribes</i>	34	<i>Pedocactus</i>	16	<i>Lithophragma</i>	8	<i>Petasites</i>	5	<i>Robinia</i>	3	<i>Tobmicea</i>	1	<i>Ozalis</i>	1
<i>Ceanothus</i>	33	<i>Sidalcea</i>	16	<i>Lygodesmia</i>	8	<i>Phlox</i>	5	<i>Sagittaria</i>	3	<i>Urtica</i>	1	<i>Ozomelis</i>	1
<i>Clarkia</i>	33	<i>Calendula</i>	15	<i>Marrubium</i>	8	<i>Pyrrocoma</i>	5	<i>Saponaria</i>	3	<i>Veratrum</i>	1	<i>Petrosedum</i>	1
<i>Origanum</i>	33	<i>Dasiphora</i>	15	<i>Papaver</i>	8	<i>Solanum</i>	5	<i>Scrophularia</i>	3	<i>Verbena</i>	1	<i>Phalaris</i>	1
<i>Spiraea</i>	33	<i>Lavandula</i>	15	<i>Ratibida</i>	8	<i>Tragopogon</i>	5	<i>Silene</i>	3	<i>Visnaga</i>	1	<i>Pilosella</i>	1
<i>Astragalus</i>	32	<i>Myosotis</i>	15	<i>Rhus</i>	8	<i>Valeriana</i>	4	<i>Sorbus</i>	3	<i>Xerophyllum</i>	1	<i>Pisum</i>	1
<i>Gilia</i>	32	<i>Rhododendron</i>	15	<i>Tagetes</i>	8	<i>Viburnum</i>	4	<i>Vitex</i>	3	<i>Zinnia</i>	1	<i>Poa</i>	1

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Hoplitia emarginata</i>	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Hoplitia fulgida</i>	0	0	0	5	10	0	0	0	0	0	0	1	1	0	0	0	0	0	21	0	0	0	8	0	1	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	50
<i>Hoplitia grinnelli</i>	0	0	3	0	1	1	0	0	0	0	7	0	4	0	0	0	0	0	3	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	18	39
<i>Hoplitia hypocrita</i>	0	1	0	0	0	0	0	0	0	2	0	0	2	0	0	0	0	0	2	1	0	1	0	1	0	0	0	0	1	0	1	0	0	0	1	0	0	0	3	16	
<i>Hoplitia lousiae</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	
<i>Hoplitia producta</i>	0	0	0	2	1	21	0	0	0	0	0	0	0	0	0	0	0	0	4	1	0	2	0	1	0	0	0	0	0	2	0	0	0	0	0	0	0	3	3	40	
<i>Hoplitia robusta</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Hoplitia sambuci</i>	0	1	1	0	0	0	0	0	0	1	0	0	16	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	0	0	11	55	
<i>Hoplitia wulatis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Hylaeus affinis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0	0	0	6	
<i>Hylaeus annulatus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Hylaeus basalis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	0	3	0	0	0	0	4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	11
<i>Hylaeus coloradensis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
<i>Hylaeus episcopalis</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	9
<i>Hylaeus granulatus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	
<i>Hylaeus leptocephalus</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	1	9
<i>Hylaeus mesilae</i>	0	0	4	1	0	5	2	0	3	0	0	0	25	0	0	0	0	26	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	9	0	0	34	111		
<i>Hylaeus modestus</i>	0	0	0	3	4	6	0	0	0	0	0	0	0	0	1	4	0	2	0	0	0	5	0	1	0	0	0	0	3	4	0	0	0	0	2	0	0	0	0	2	37
<i>Hylaeus nevadensis</i>	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	25
<i>Hylaeus punctatus</i>	0	0	0	0	14	7	0	0	0	0	0	0	0	0	1	3	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	27
<i>Hylaeus rubriclavis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	72	82	
<i>Hylaeus verticalis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	3	
<i>Hylaeus woottoni</i>	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	14
<i>Lasioglossum (Lasioglossum)</i>	0	0	0	0	1	57	0	0	0	0	0	0	1	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	72	
<i>Lasioglossum albipenne</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	
<i>Lasioglossum albobirtum</i>	0	0	1	0	0	0	0	0	1	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	5	14	
<i>Lasioglossum allonotum</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	6
<i>Lasioglossum anhypops</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	4	11
<i>Lasioglossum aspilurus</i>	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	11
<i>Lasioglossum athabascense</i>	0	0	0	0	5	0	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	9
<i>Lasioglossum brunneiventre</i>	0	0	1	0	0	0	0	0	2	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7
<i>Lasioglossum buccale</i>	0	0	0	0	2	2	0	0	0	0	0	0	6	0	0	6	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	18
<i>Lasioglossum colatum</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	
<i>Lasioglossum cooleyi</i>	0	4	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	20
<i>Lasioglossum cordleyi</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Lasioglossum cressonii</i>	0	0	0	0	16	9	0	0	1	0	0	0	0	0	0	5	0	1	1	0	0	0	2	0	0	0	1	2	4	0	1	0	0	0	0	0	3	1	47		
<i>Lasioglossum dashwoodi</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	
<i>Lasioglossum diversopunctatum</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	3	
<i>Lasioglossum egregium</i>	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	10	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	19	
<i>Lasioglossum fozi</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	12	
<i>Lasioglossum glabriventre</i>	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	4	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	8	
<i>Lasioglossum helianthi</i>	0	0	0	0	0	3	0	0	0	0	1	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	13	24	
<i>Lasioglossum hyalinum</i>	0	0	0	0	0	0	0	0	2	0	0	0	3	0	0	0	0	6	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	17		
<i>Lasioglossum incompletum</i>	0	0	0	0	18	8	1	0	1	0	0	0	3	0	0	0	0	17	2	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	25	78		
<i>Lasioglossum inconditum</i>	0	0	0	6	20	0	0	0	1	0	0	2	2	0	0	0	1	14	0	0	0	0	0	2	2	0	0	0	1	0	0	0	0	0	0	3	16	70			
<i>Lasioglossum kincaidii</i>	0	0	0	4	3	7	0	0	0	0	0	0	9	0	0	2	1	0	0	0	1	0	0	0																	

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL		
<i>Lasiosglossum titusi</i>	0	0	0	1	9	137	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	14	0	168		
<i>Lasiosglossum trizonatum</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6		
<i>Lasiosglossum villosulum</i>	0	0	0	0	7	65	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	7	0	3	0	0	20	0	0	0	0	1	105		
<i>Lasiosglossum yukonae</i>	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6		
<i>Lasiosglossum zephyrum</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0	7		
<i>Lasiosglossum zonulum</i>	0	0	0	0	13	3	0	0	0	0	0	0	0	0	0	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	0	1	0	0	0	27		
<i>Lasiosglossum zonulus</i>	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4		
<i>Megachile angelarum</i>	0	0	2	0	5	30	0	0	0	0	0	2	0	9	0	1	7	2	4	0	0	0	0	1	0	2	0	0	0	0	3	0	1	0	0	0	0	0	0	0	5	74
<i>Megachile apicalis</i>	0	0	3	0	0	0	1	0	1	0	6	0	23	0	0	0	0	0	11	1	0	1	0	2	0	0	0	0	0	1	0	1	1	0	0	11	0	4	10	77		
<i>Megachile brevis</i>	0	1	2	1	3	6	0	0	0	0	0	0	10	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	31		
<i>Megachile centuncularis</i>	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4		
<i>Megachile coquillettii</i>	0	0	0	0	0	0	0	0	1	0	0	0	6	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	11	
<i>Megachile fidelis</i>	0	0	4	0	1	8	0	1	0	0	0	0	0	0	0	4	4	1	5	5	0	0	0	1	0	2	0	0	0	0	0	0	0	0	1	0	0	0	0	6	43	
<i>Megachile frigida</i>	0	0	0	0	15	4	0	0	0	0	0	0	0	0	0	0	3	0	1	0	0	0	0	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	29	
<i>Megachile gemula</i>	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	6	
<i>Megachile gravata</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile lapponica</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	2	4
<i>Megachile lippiae</i>	0	0	2	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	9	
<i>Megachile melanophaea</i>	0	0	0	0	7	0	0	0	0	1	0	1	0	0	0	1	3	0	3	0	0	0	0	4	0	0	0	1	1	0	0	0	0	0	4	0	0	1	0	1	28	
<i>Megachile mellitarsis</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile mendica</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile montivaga</i>	0	0	1	1	12	9	0	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	2	0	5	0	0	0	0	0	0	0	0	0	0	3	0	1	1	40	
<i>Megachile nevadensis</i>	2	0	15	1	0	0	0	0	3	0	10	0	43	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	110	
<i>Megachile onobrychidis</i>	0	0	3	0	0	0	0	0	3	1	5	0	67	0	0	0	0	0	14	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	16	113	
<i>Megachile parallela</i>	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	2	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15	
<i>Megachile perthirta</i>	0	4	3	2	70	76	0	3	1	4	1	9	23	3	1	2	24	8	15	2	0	1	0	4	0	2	0	3	2	0	0	5	0	0	0	1	3	3	17	292		
<i>Megachile pugnata</i>	0	0	3	3	13	2	0	1	1	0	2	0	0	0	0	0	0	0	6	2	2	0	0	4	0	2	0	0	0	0	0	0	0	0	0	1	0	0	10	52		
<i>Megachile relativa</i>	0	0	0	0	13	0	0	0	0	1	0	0	0	0	0	0	1	0	16	0	0	0	0	2	0	4	0	0	0	0	1	0	0	0	0	1	0	0	2	41		
<i>Megachile rotundata</i>	0	0	11	0	0	19	0	1	0	1	5	0	28	0	0	0	6	0	3	0	0	0	0	2	0	0	0	0	0	0	0	1	0	0	0	3	0	0	3	83		
<i>Megachile subnigra</i>	0	0	1	0	0	0	0	0	2	0	0	0	6	0	0	0	0	0	11	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	31	
<i>Megachile tezana</i>	0	0	0	0	0	4	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	17	
<i>Megachile umatillensis</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Megachile wheeleri</i>	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	12	
<i>Melecta pacifica</i>	0	0	0	0	0	0	1	0	0	0	0	0	1	12	0	0	0	0	6	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	24
<i>Melecta separata</i>	0	0	0	1	0	0	0	0	0	0	0	0	3	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	13		
<i>Melecta thoracica</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	
<i>Melissodes agilis</i>	0	13	2	0	0	0	2	0	0	2	0	0	5	61	0	0	0	0	0	3	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	9	9	108	
<i>Melissodes binatrix</i>	0	4	12	0	0	0	0	0	0	0	6	3	37	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	9	75		
<i>Melissodes communis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Melissodes dagosus</i>	0	0	0	0	0	0	0	0	0	0	0	0	24	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25		
<i>Melissodes glenwoodensis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Melissodes lupinus</i>	0	0	0	0	0	48	0	0	0	0	0	0	1	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	62		
<i>Melissodes lustrus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	101	107		
<i>Melissodes menuchus</i>	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9		
<i>Melissodes metenus</i>	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	11		
<i>Melissodes microstictus</i>	0	1	0	0	41	62	0	0	0	0	0																															

	Adams	Astin	Benton	Chelan	Columbia	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Skamania	Stromholm	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Osmia cobaltina</i>	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	
<i>Osmia coloradensis</i>	0	0	0	9	1	0	0	0	1	1	0	1	1	0	0	1	1	2	11	0	0	1	0	7	4	4	0	22	1	0	0	3	0	0	0	0	1	1	3	15	67
<i>Osmia cornifrons</i>	0	0	0	0	0	6	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	
<i>Osmia cyaneella</i>	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Osmia densa</i>	0	0	0	5	1	0	0	0	0	1	0	0	0	0	0	0	0	8	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	23
<i>Osmia ednae</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	
<i>Osmia esigua</i>	0	0	1	2	1	1	0	0	0	0	0	2	3	0	0	2	0	0	10	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	11	35
<i>Osmia gilliarum</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia grinnelli</i>	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	24	
<i>Osmia integra</i>	0	0	0	0	0	0	0	0	0	0	4	0	29	0	0	0	0	0	8	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10	58	
<i>Osmia inurbana</i>	0	0	0	0	0	0	0	1	1	2	0	1	0	0	0	0	0	5	2	0	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	25	
<i>Osmia iridis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Osmia justa</i>	0	0	0	2	6	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia lacta</i>	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia lignaria</i>	0	0	0	1	4	6	0	0	0	0	0	0	3	0	0	3	2	0	4	1	0	0	3	0	0	0	0	0	0	0	6	1	0	0	0	0	1	0	5	40	
<i>Osmia longula</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	7		
<i>Osmia malina</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Osmia montana</i>	0	0	0	10	0	0	0	0	0	0	0	1	2	0	0	0	0	22	9	0	1	0	27	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	26	101	
<i>Osmia nemoris</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Osmia nifoata</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	5		
<i>Osmia nigricornis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Osmia odontogaster</i>	0	0	0	1	4	0	0	0	0	0	0	3	0	0	0	0	0	16	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	32	
<i>Osmia pentastemonis</i>	0	0	0	11	1	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	3	3	4	4	0	1	1	0	0	0	0	0	0	0	0	0	4	33	
<i>Osmia phacelia</i>	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia prozima</i>	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	1	9	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	1	1	19		
<i>Osmia pusilla</i>	0	0	0	5	5	10	0	0	0	0	0	0	0	0	0	0	0	27	0	0	0	3	0	1	0	0	0	0	0	0	0	0	0	0	6	0	1	1	59		
<i>Osmia rasilini</i>	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	7		
<i>Osmia regulina</i>	0	0	0	1	0	0	0	0	0	0	5	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	0	2	25			
<i>Osmia sculleni</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6		
<i>Osmia sodula</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12		
<i>Osmia subaustriaca</i>	0	0	0	2	0	0	0	0	0	0	0	0	2	0	0	0	0	6	0	0	0	0	4	0	0	0	0	0	0	0	1	0	0	0	0	0	0	4	19		
<i>Osmia tezana</i>	0	0	0	0	4	0	0	0	0	0	0	0	1	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	11		
<i>Osmia trifoliaria</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1			
<i>Osmia tristella</i>	0	0	0	7	6	0	0	0	0	0	0	1	0	0	0	0	5	0	17	0	0	0	0	1	0	16	0	0	2	0	0	0	0	0	0	1	0	30	86		
<i>Osmia vandykei</i>	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15		
<i>Panurginus atriceps</i>	0	0	0	11	45	9	0	0	0	0	0	2	0	0	0	10	1	35	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	8	4	126		
<i>Panurginus inepitus</i>	0	0	0	5	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7		
<i>Perdita claypolei</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	11		
<i>Perdita lingualis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2		
<i>Perdita nevadensis</i>	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	24		
<i>Perdita nuda</i>	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Perdita oregonensis</i>	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5		
<i>Perdita salicis</i>	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Perdita wyomingensis</i>	0	0	0	1	0	0	8	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13		
<i>Perdita zonalis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Protosmia rubifloris</i>	0	0	0	1	0	6	0	0	0	0	0	0	1	0	0	0	0	1	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	16		
<i>Pseudanthidium nanum</i>	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9		
<i>Sphecodes pecosensis</i>	0	0	0	0	3	5	0	0	0	0	0	1</																													

Table 5: Your determination accuracy in 2024.

Taxon
No specimens identified

7 Taxonomic Accuracy, 2024

In 2024, you identified 0 of your 3 specimens to genus level and 0 to species level (see Table 5). In total, volunteers from the Washington Bee Atlas project identified 1.3 % (127) of the 10006 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (see Table 6).

Table 6: Determination accuracy for all volunteers in 2024.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN

Table 7: Your determination accuracy.

Taxon
No specimens identified

8 Taxonomic Accuracy, All Years

Over your time in the Atlas you identified 0 of your 3 specimens to genus level and 0 to species level (see Table 7). In total, volunteers from the Washington Bee Atlas project identified 0.6 % (127) of the 22418 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (See Table 8).

Table 8: Determination accuracy for all volunteers.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN